

Student's name: Chen Povsavortey

Faculty of Digital Economy

Class: AFT

Course: Blockchain

SECP256k1 Elliptic Curve: A Literature Review

1. Introduction

Blockchain technology depends strongly on cryptography to keep transactions secure and trustworthy. Every time a user sends cryptocurrency, cryptographic techniques are used to prove ownership and prevent fraud. One of the most important cryptographic tools used in blockchain systems, especially Bitcoin, is elliptic curve cryptography (ECC).

Among many elliptic curves, **SECP256k1** is the curve chosen by Bitcoin for generating digital signatures. This literature review discusses what SECP256k1 is, how it works at a basic level, why it is used in blockchain systems, and how secure and efficient it is. In addition, this paper reviews an IEEE research study that analyzes the implementation of SECP256k1 in cryptographic systems.

2. Definition of SECP256k1

SECP256k1 is a specific elliptic curve used in public-key cryptography. It was defined by the Standards for Efficient Cryptography Group (SECG) and is mainly used in the Elliptic Curve Digital Signature Algorithm (ECDSA). The curve works over a 256-bit finite field, which provides strong security while remaining efficient.

In blockchain systems such as Bitcoin, SECP256k1 is used to generate a pair of cryptographic keys: a private key and a public key. The private key is kept secret by the user, while the public key is shared with others to verify digital signatures. This process allows users to prove ownership of funds and authorize transactions without exposing their private keys

3. Mathematical Foundation of SECP256k1

Elliptic curve cryptography is based on mathematical equations that describe points on a curve. In ECC, a private key is generated as a random number, and the public key is created by performing a mathematical operation on the elliptic curve using that private key. While generating the public key is easy, reversing the process to find the private key is extremely difficult. This difficulty is known as the Elliptic Curve Discrete Logarithm Problem, which provides the core security of ECC.

4. Use of SECP256k1 in Bitcoin and Blockchain Systems

Bitcoin uses SECP256k1 to secure transactions through digital signatures. When a user creates a Bitcoin transaction, it is signed using ECDSA with SECP256k1. Other participants in the network can then verify the signature using the public key, ensuring that the transaction is valid and has not been modified.

One main reason Bitcoin uses SECP256k1 is efficiency. Compared to older cryptographic methods like RSA, elliptic curve cryptography provides the same level of security with much smaller key sizes. This reduces computation time and storage requirements, which is important in decentralized systems with many users and transactions.

According to Antonopoulos, elliptic curve cryptography allows Bitcoin to remain secure while keeping transaction verification fast and practical for global use.

5. Security and Efficiency of SECP256k1

SECP256k1 offers strong security because of its large key space and reliance on complex mathematical problems that are difficult to solve. Brute-force attacks are practically impossible due to the enormous number of possible private keys.

In terms of efficiency, SECP256k1 is faster than some other elliptic curves because of its simpler structure. This makes it suitable for blockchain systems that require frequent cryptographic operations. However, the security of SECP256k1 also depends on correct implementation. Poor random number generation or reused nonces can introduce vulnerabilities, even if the curve itself is secure.

6. Review of Research Paper

This literature review examines the IEEE research paper titled “**Implementation of Elliptic Curve Cryptosystem with Bitcoin Curves on SECP256k1.**” The main goal of the paper is

to study how elliptic curve cryptography, particularly SECP256k1, is implemented and how it performs in real-world systems such as Bitcoin.

The authors explain how SECP256k1 is used within the Elliptic Curve Digital Signature Algorithm to secure digital communications. The paper also discusses potential weaknesses related to poor nonce generation and shows how these issues can affect security if not handled correctly. Through experimental analysis, the study demonstrates that SECP256k1 is efficient and reliable when proper cryptographic practices are followed.

This research is important because it not only supports the effectiveness of SECP256k1 but also highlights the need for careful implementation. It helps developers and researchers better understand how to maintain strong security in blockchain systems.

7. Conclusion

SECP256k1 is a key component of modern blockchain technology, especially in Bitcoin. Its use of elliptic curve cryptography allows secure digital signatures with high efficiency and smaller key sizes. The reviewed IEEE research paper confirms that SECP256k1 performs well in practical implementations while emphasizing the importance of secure nonce generation.

Overall, SECP256k1 remains a reliable and widely used elliptic curve that supports the security, integrity, and trust required in blockchain networks.

REFERENCES:

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